An Interactive VR System for Visualizing Traditional Chinese Medicine Facial Diagnostics Using Multimodal AI

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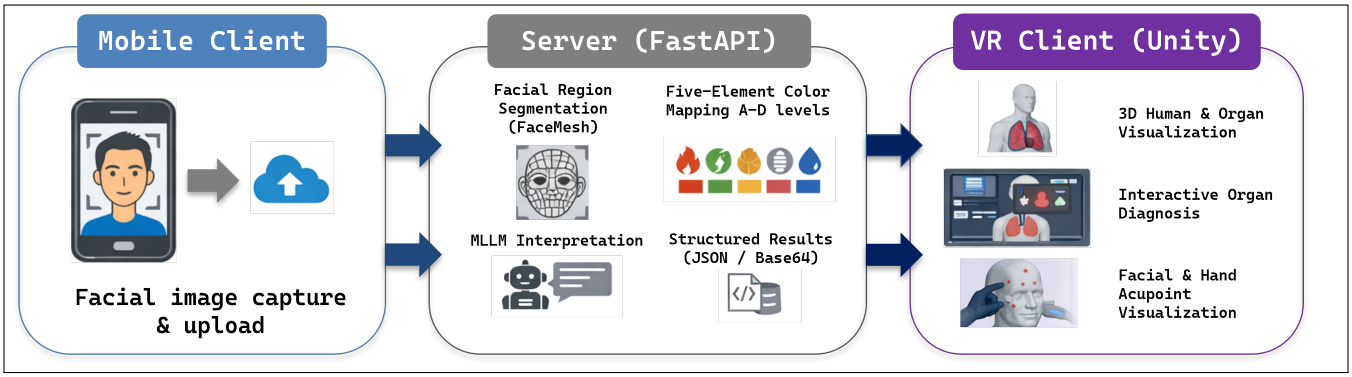

Figure 1: The workflow of Traditional Chinese Medicine Facial Diagnostics.

Abstract

This project integrates virtual reality (VR), multimodal large language models (MLLMs), and image processing to create an immersive interactive system for visualizing the principles of Traditional Chinese Medicine (TCM) facial diagnosis. In TCM theory, specific facial regions correspond to the Five Zang Organs and the Five Elements, forming a symbolic mapping that reflects internal physiological states. The forehead, glabella, nose bridge, cheeks, and chin are traditionally associated with the heart, liver, spleen, lungs, and kidneys, respectively. However, TCM knowledge is often conveyed through abstract and non-interactive formats. By leveraging VR, this work reconstructs and visualizes TCM facial diagnosis in a spatial and experiential manner, enabling users to explore its theoretical and cultural foundations through immersive interaction.

Keywords

Traditional Chinese Medicine (TCM), Virtual Reality, Facial Visualization, Multimodal Large Language Models, Immersive Interaction

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1 Introduction

Facial diagnosis is an important component of Traditional Chinese Medicine (TCM), where observations of color, brightness, and changes in different regions of the face are used to infer the general condition of the five zang organs. Classical TCM theory states that the forehead corresponds to the heart, the area between the eyebrows to the liver, the nose bridge to the spleen, the cheeks to the lungs, and the chin to the kidneys. These facial regions are also associated with the Five Elements and their representative colors—for example, the heart corresponds to fire and the color red, while the liver corresponds to wood and greenish hues. Together, these mappings form an interpretive framework in which external facial features symbolically reflect internal physiological tendencies. However, traditional TCM knowledge is usually presented through text descriptions or static diagrams, which are often abstract and difficult for learners to understand intuitively.

With the development of virtual reality (VR), image processing, and artificial intelligence (AI), researchers have begun to explore how immersive environments can support medical and health-related learning. Prior studies have shown that VR can effectively visualize physiological conditions or sensory impairments in ways that enhance user understanding. For example, Orlosky et al. developed an eye-tracked VR/AR simulation suite that recreates various visual deficits to help learners better experience vision-related difficulties, demonstrating VR's potential in medical education [4]. Similarly, Alexander et al. showed how VR simulations can support

experience-based learning in physical diagnosis training [1]. These works highlight VR's advantages in transforming abstract medical concepts into intuitive visual experiences.
At the same time, digitalization efforts in TCM have made notable progress. Li et al. proposed a computer-assisted diagnosis system based on facial image analysis, demonstrating the feasibility of extracting meaningful features from facial images for TCM-related applications [3]. Hou et al. reviewed recent advances in artificial intelligence for traditional medicine, including facial feature modeling, knowledge representation, and multimodal learning [2]. These studies provide a technological foundation for modernizing the presentation of traditional diagnostic concepts.
Building on these developments, we designed an interactive TCM facial diagnosis system that integrates VR, large language models (LLMs), and image processing. Users upload a facial image via a mobile device, and the system automatically identifies facial regions, analyzes color features, and generates explanatory text related to the Five Zang Organs and the Five Elements. In VR, users can explore organ–region relationships, observe color mappings, and further examine representative facial acupoints to develop a more intuitive understanding of TCM facial diagnosis. The goal of this project is not medical diagnosis, but to use modern technology to make TCM knowledge more visual, contextual, and accessible through immersive cultural learning experiences.

2 System Overview

The system consists of three components—the mobile client, the server, and the VR client—connected via Internet to form a complete workflow from image acquisition to immersive visualization.

2.1 Mobile Client

The mobile client serves as the entry point of the system, responsible for capturing the user's facial image and uploading it to the server for analysis. After the user takes or selects a photo, the application performs basic preprocessing and sends it to the server through an analysis request. Once the server returns the results—including the estimated organ states, corresponding Five-Element color intensities, and the annotated facial-region visualization—the mobile client displays these outputs for user confirmation. Acting as both the data input module and the result preview interface, the mobile client initiates and concludes the interaction flow.

2.2 Server

The server serves as the core computational module of the system, responsible for facial-region segmentation, Five-Element–based color analysis, and multimodal large language model (MLLM) interpretation. Its workflow is divided into two main components, described below.

2.2.1 Facial Region Segmentation and Five-Element Color Mapping. Upon receiving the uploaded facial image, the server first compresses it to half of its original width and height to reduce transmission and computation overhead. The system then applies OpenCV's Haar face detector to obtain a coarse facial bounding box, followed by MediaPipe FaceMesh to extract 468 detailed facial landmarks. These landmarks enable precise localization of the five TCM-related facial regions: the forehead (heart), glabella (liver), nose bridge (spleen), cheeks (lung), and chin (kidney).
For each region, the server calculates the average RGB color and compares it with the predefined Five-Element color mapping: heart–fire–red, liver–wood–green, spleen–earth–yellow, lung–metal–white, and kidney–water–black. Each organ is assigned four template color levels (A–D), and the server determines the closest level based on color distance, producing a symbolic representation of the organ's state.
To support VR visualization, the server also overlays semi-transparent color masks on the corresponding facial regions—using FaceMesh landmark geometry—to generate a "color-annotated" facial image. This annotated image, together with the organ-level color classifications, forms the basis of the system's visual output.

Table 1: Mapping of Five Organs, Facial Regions, Five Elements, and Color Levels

Organ	Facial Region	Five Element	Color Levels (A → D)
Heart	Forehead	Fire	A: #F4A3A3 → B: #F26A6A → C: #D83434 → D: #9C1C1C
Liver	Glabella	Wood	A: #A8D5BA → B: #6BBF59 → C: #3F9142 → D: #1E5631
Spleen	Nose Bridge	Earth	A: #F7E3A3 → B: #F2CC5C → C: #E0A800 → D: #B58100
Lung	Cheeks	Metal	A: #F9F9F9 → B: #E5E5E5 → C: #CCCCCC → D: #B3B3B3
Kidney	Chin	Water	A: #A0A4B8 → B: #6B7280 → C: #374151 → D: #111827

2.2.2 MLLM-Based Interpretation Using Qwen. After obtaining the segmented facial regions, the server invokes the multimodal large language model Qwen-VL to generate descriptive explanations related to TCM facial diagnosis. The compressed facial image is encoded in base64 and passed to the model along with a tailored prompt instructing it to: 1) interpret the image in alignment with the five-organ facial mapping; 2) provide observation-based descriptions focusing on color and brightness;
Qwen-VL returns a paragraph that discusses the face from the perspectives of the heart, liver, spleen, lung, and kidney. The server then parses the output and separates the text into organ-specific segments, enabling flexible presentation on the VR client.
Finally, the server packages all results—including the organ-level classifications, Five-Element color intensities, the annotated image, the full Qwen-generated description, and the organ-specific text slices—into a unified structured format. These results are returned both to the mobile client for immediate verification and to the VR client for immersive three-dimensional visualization, completing the system's analysis and output pipeline.

2.3 VR Client

The VR client serves as the primary immersive interface of the system, responsible for integrating the server's analytical results, Five-Element color mappings, and the 3D anatomical model into an interactive spatial experience. In the VR environment, three components form the core of the visual presentation: (1) the annotated facial images returned by the server, (2) a Unity-based 3D human model including the head and internal organs, and (3) an interactive panel for the five-organ diagnostic results.

2.3.1 Visualization of Annotated Facial Images. The VR client first loads two types of image data provided by the server: 1) the compressed original facial image, and 2)the Five-Element–colored overlay image based on FaceMesh segmentation.

2.3.2 Unity-Based 3D Human Model and Organ Visualization. To strengthen the conceptual link between facial regions, internal organs, and Five-Element attributes, a full 3D human model—including head structures and the five organs (heart, liver, spleen, lung, kidney)—was built in Unity.

Upon receiving the server's color-level classifications (A–D) and the corresponding Five-Element color values: 1) each organ in the 3D model is highlighted using its elemental color (e.g., red for the heart, green for the liver); 2) the organ's internal state is represented through different color intensities, where milder hues correspond to balanced states and deeper, more saturated hues indicate stronger deviations.

This design allows users to experience organ states in an immersive, visually intuitive way.

2.3.3 Interactive Five-Organ Diagnostic Panel. The five diagnostic results returned by the server are presented in VR as five interactive buttons, each styled with its Five-Element color. When the user selects a button through VR gesture interaction: 1) the button is highlighted using its corresponding elemental color; 2) the matching organ in the 3D anatomical model becomes visually activated and rendered according to its resulting color level; 3) the AI-generated explanation (from Qwen) is displayed, linking visual cues with TCM facial-diagnosis interpretations.

This interaction loop enables users to explore the system's symbolic diagnostic outcomes in a vivid and intuitive manner.

2.3.4 3D Acupoint Visualization and Therapeutic Suggestions. To enrich the educational value of the system, representative acupoints associated with each organ are included:

(a) **Facial acupoints on the 3D human model**
Key acupoints on the head—such as Yintang, Yingxiang, Taiyang, Chengjiang, etc.—are displayed as markers, helping users understand the connection between facial features, organs, and classical meridian theory.

(b) **Hand acupoints triggered by downward head movement**
When the user looks downward in VR, a secondary panel reveals hand acupoints commonly associated with organ regulation, e.g. Heart → Shenmen, Liver → Taichong.

These acupoint recommendations allow users to intuitively access TCM health-preservation knowledge without requiring physical controllers.

3 Conclusion And Discussion

This paper presents an immersive VR-based system that integrates facial image analysis, multimodal AI, and traditional Chinese medicine theory to support intuitive exploration of facial diagnosis principles. By visualizing facial regions, organ mappings, Five-Element color relationships, and representative acupoints in a 3D interactive environment, the system demonstrates how VR can effectively support experiential learning of culturally grounded knowledge. While the current implementation focuses on symbolic interpretation rather than medical diagnosis, it highlights the potential of combining AI and VR for educational visualization. Future work will include user studies, interaction refinement, and the expansion of additional TCM knowledge modules.

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